Creating an LLVM Runtime

EuroLLVM 2026

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Outline

Introduction

Let's Create a Runtime

And They Lived Happily...

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Outline

Introduction
- Why This Tutorial?
- History
- CMake Basics
- Rationale
- Goal

Let's Create a Runtime

And They Lived Happily...

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Why This Tutorial

[redacted] 1 hour ago Member

After this change, it's no longer possible to run openmp tests in a non-bootstrapping build (i.e. without building clang at the same time).

Previously it was possible to run the tests in a standalone build, by building the `FileCheck` and `not` binaries and pointing at them with `OPENMP_LLVM_TOOLS_DIR` - like I did at mstorsjo/llvm-mingw@ `20260210` /.github/workflows/build.yml#L583-L612.

Now after this change (or before, when doing a runtimes build), running tests only works if there's a `FileCheck` target already in the build, i.e. if doing a bootstrapping build:

llvm-project/openmp/cmake/OpenMPTesting.cmake
Lines 52 to 74 in c6425aa

```
52    if (${OPENMP_STANDALONE_BUILD})
53      find_standalone_test_dependencies()
54
55      # Set lit arguments.
56      set(DEFAULT_LIT_ARGS "-sv --show-unsupported --show-xfail")
57      if (MSVC OR XCODE)
58        set(DEFAULT_LIT_ARGS "${DEFAULT_LIT_ARGS} --no-progress-bar")
59      endif()
60      if ("${CMAKE_SYSTEM_NAME}" MATCHES "AIX")
61        set(DEFAULT_LIT_ARGS "${DEFAULT_LIT_ARGS} --time-tests --timeout=3000")
62      endif()
63      set(OPENMP_LIT_ARGS "${DEFAULT_LIT_ARGS}" CACHE STRING "Options for lit.")
```


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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

Yep, just for context from the Rust side. We have a lot of builders in CI, but the slowest one we have is the x86 dist builder, which we use to e.g. create compiler artifacts. We already build LLVM there around 4 times due to PGO and Bolt, so we really try hard not to make it any slower, since we are already at ~3.5 hrs, and we run this CI before any merge into main.

One of the compromises there is that we have two checkouts of LLVM - once as a standalone llvm/llvm-project build (A) and once as a submodule of rustc in `src/tools/llvm-project`. We use build A to compile C/C++ code, and keep rebuilding the second checkout (B) during bootstrapping.
We only build clang as part of our LLVM Build A. However, rustc is based on LLVM B, and as such we'd also want the openmp and offload libraries to also be part of our LLVM Build B, which does not build the in-tree clang to save compile times.

We've tried for quite a bit to get the runtimes to build despite no in-tree clang or lld in LLVM B, and failed, I'm happy to share all the issues I ran into there. A standalone build instead allows me to use clang from build A to build offload+openmp from checkout B, that's compatible with what we have in rustc. So until that is fixed, standalone builds are likely the only builds that our infra team would be ok with.

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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

[redacted] 1 hour ago • edited Contributor

We were using the OpenMP standalone build mode in order to build the static libomp.a runtime library which cannot be build together with the shared libomp.so (there is no FLANG_RT_ENABLE_SHARED and FLANG_RT_ENABLE_STATIC equivalent for libomp, the LIBOMP_ENABLE_SHARED flag is binary: one can get either shared or static library, but never both in the same build).

The OpenMP standalone build mode has been removed by llvm/llvm-project#182022

This complicates our situation. Considering almost nonexistent demand for the static OpenMP runtime library, it is not worth pursuing it.

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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

Contributor

[redacted] 1 hour ago • edited

Member

[redacted] on Jan 7 • edited

That comes with two drawbacks: 1) checking out `llvm` alone (I'm not talking about `llvm-project`) takes a lot of time. 2) the way to pass CMake arguments to runtime build is extremely user unfriendly. If these two can't be improved, I strongly object the removal of standalone build.

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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

[redacted] 1 hour ago • edited Member

[redacted] on Dec 8, 2025

Rust colleagues rely on standalone builds for their offloading extensions (see rust-lang/rust#148671).

CC @ZuseZ4

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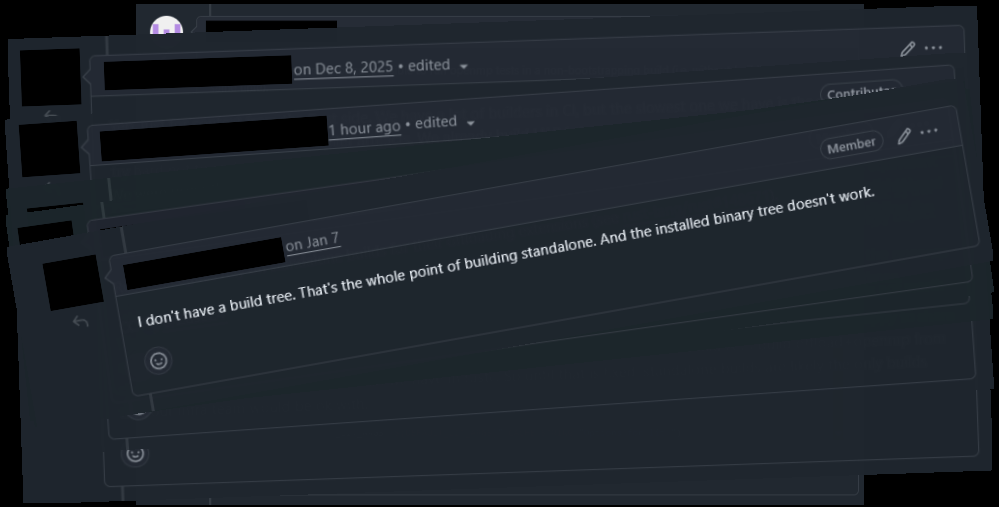
Why This Tutorial



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Why This Tutorial

[redacted] on Dec 8, 2025 • edited

Contributor

[redacted] on Jul 26, 2025

Member

It is definitely not the same!
For standalone build, I used to not needing the llvm-project tree. I could build libomp by just downloading the OpenMP release tarball: `llvmorg-21.1.0-rc1` openmp-21.1.0-rc1.src.tar.xz (download)

I just realized that this workflow got bricked by #62240

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History

- Initially, each subproject had its own SVN repository
- No difference between projects and runtimes
- Step 1: `LLVM_ENABLE_PROJECTS`
- Step 2: Sort into tools/<subproject> and runtimes/<subproject> directories
- Step 3: `LLVM_ENABLE_RUNTIMES`
- `https://github.com/llvm/llvm-project/pull/170693#issuecomment-3720262808`
- Original proposal: `https://lists.llvm.org/pipermail/llvm-dev/2020-October/145997.html`

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CMake Basics

- Do not use `set(VAR_NAME)`

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CMake Basics

- Do not use `set(VAR_NAME)`
- Do not use `if (${VAR_NAME})`

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Rationale

- Compile runtimes using just-built Clang
- Compile runtimes for each target platform
  - Cross-compilation
- Avoid combinatorial explosion of build modes

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Purpose

Our Runtime

```
void _dublin_hello_world() {
  puts("Dia daoibh, a dhomhain!\n");
}
```

- libdublin.a/.so
- Linux Only
  - No Windows/MacOS/AIX/...

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Outline

Introduction

Let's Create a Runtime
- Step 0: Setup
- Step 1: Register with the LLVM Build System
- Step 2: Build a Library Artifact
- Step 3
- Step 4
- Step 5
- Step 6
- Step 7
- Step 8
- Step 9
- Step 10
- Step 11
- Step 12
- Step 13
- Step 14
- Step 15

And They Lived Happily...

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Setup

- Code found at `https://github.com/meinersbur/llvm-dublin`
- One branch per step
  - Starting point is `https://github.com/Meinersbur/llvm-dublin/tree/dublin-0-main`
- Changes as PR to previous branch

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Step 1: Register with the LLVM Build System

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Step 2: Build a Library Artifact

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Add an Example that uses the Library

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Build Modes

- Bootstrapping runtimes
- Runtimes standalone
- Runtimes standalone with an LLVM install dir
- Runtimes standalone without an LLVM build

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Artifact Output Location

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Shared and Static Libraries

- Add a Configuration option
-

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Installation

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Add a Regression Test

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Add a Unittest

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Sphinx Docs

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Doxygen Docs

▶

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Cross-Compilation

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Depending on Other LLVM Libraries

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Using C++ in a Runtime

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Accelerator Offloading

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Outline

Introduction

Let's Create a Runtime

And They Lived Happily...
- More Topics
- Build System Improvements
- Conclusion

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Uncovered Topics

- Platform Support: Windows/MacOS/AIX/Fuchsia/...
- Multilib
- ...

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Future Improvements

- Inform LLVM developers and distributions

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- `LLVM_ENABLE_RUNTIMES` Documentation

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- `LLVM_ENABLE_RUNTIMES` Documentation
- Propagate `RUNTIMES_*` options

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  - Use `libdublin` as a template?

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- ...

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Conclusion

- Now you know how `LLVM_ENABLE_RUNTIMES` works
- Use the Knowledge Wisely
- I currently cannot recommend working on improving the system, waste of your time

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